

## 100 Days of Azure DevOps

## Day 55 Handout — Learn the airport (Pod, Deployment, Service, namespace) before you land a lemonade stand on AKS

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

### Architecture A — request path (kube-system is not your app)

| Object      | Job                                                                | Where it lives                                                                                      |
|-------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Namespace   | Tenancy lite. DNS and RBAC boundary                                | workloads for the app. kube-system for CoreDNS, kube-proxy, connectivity — not Deployments of myapp |
| Pod         | Smallest run unit. One or more containers, shared netns            | Created by a controller. Naked Pods in prod are pets                                                |
| Deployment  | replicas + selector + pod template. Rolling updates                | Workloads namespace. selector.matchLabels must equal template labels                                |
| Service     | Stable virtual IP / DNS. Selects Pods by label, not by folklore IP | ClusterIP for in-cluster. containerPort must match the process                                      |
| kube-system | Cluster plumbing. CoreDNS answers Service DNS                      | kubectl get pods -n kube-system is literacy. Do not kubectl apply your app there                    |

### Architecture B — decide cluster today: kind vs reading YAML vs AKS

| Constraint                  | Default                                                     | Move when                                                                     |
|-----------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------|
| Need kubectl apply practice | kind or minikube on the personal PC                         | You cannot install a local cluster — still write YAML and trace Service → Pod |
| Need Azure control plane    | Not today. AKS is Day 56 and optional                       | You already accepted a destroy date this weekend                              |
| Image tag                   | Pin the ACR digest or Build.BuildId in the manifest         | image: ...:latest is a mood, not a deploy                                     |
| Namespace                   | workloads (create it). Not default-forever, not kube-system | default is acceptable only as a 10-minute kind demo                           |
| Service type                | ClusterIP + port-forward for kind                           | LoadBalancer on kind is a sketch; AKS LoadBalancer is a bill                  |

### Architecture C — pitfalls that schedule nothing and look like Kubernetes is broken

| Pitfall                   | What you see                                            | Fix                                                                       |
|---------------------------|---------------------------------------------------------|---------------------------------------------------------------------------|
| Label mismatch            | Deployment availableReplicas 0. Service endpoints empty | template.metadata.labels == spec.selector.matchLabels == Service.selector |
| App in kube-system        | Your crashLoop next to CoreDNS. RBAC horror later       | namespace: workloads. Leave kube-system to the cluster                    |
| Port mismatch             | Endpoints exist, curl hangs                             | containerPort and Service targetPort = listen port                        |
| :latest + imagePullPolicy | Old bits keep running; you swear you deployed           | Pin tag/digest; set imagePullPolicy IfNotPresent vs Always on purpose     |
| No resources              | Fine on kind; HPA later stares at zero requests         | Set requests even on the hello Deployment — Day 59 will need them         |

### One-liner to remember

**Kubernetes is an airport for containers — powerful, expensive, and overkill for a lemonade stand. Learn Pod / Deployment / Service / namespace. Do not land on AKS until the stand needs runways.**

## Step-by-step lab (personal Azure / Azure DevOps only)

Personal PC. Prefer kind or minikube. Do not create AKS today. Personal repo k8s/ manifests.

- 1 Write k8s/namespace.yaml (workloads), k8s/deployment.yaml, k8s/service.yaml for the sample app. Image from ACR with a pinned tag, not latest. replicas: 1. resources.requests set (cpu 50m, memory 64Mi).
- 2 Confirm labels: Deployment selector, pod template, Service selector are identical (app: myapp). Confirm containerPort matches the process.
- 3 If kind/minikube exists: kubectl apply -f k8s/ && kubectl -n workloads get deploy,po,svc,ep. kubectl -n kube-system get po — look, do not apply the app there.
- 4 If no local cluster: still write the YAML. Trace a request: Service ClusterIP → endpoints → Pod IP:containerPort. That tracing is the lab.
- 5 kubectl explain deployment.spec.template.spec.containers (or read the YAML) until you can say why a naked Pod is the wrong object.
- 6 Write docs/k8s-day55.md: the three objects, why kube-system is off-limits for myapp, and that AKS is optional tomorrow with a destroy date.

## Starter manifests — workloads namespace, not kube-system

```
apiVersion: v1
kind: Namespace
metadata: { name: workloads }
---
apiVersion: apps/v1
kind: Deployment
metadata: { name: myapp, namespace: workloads }
spec:
  replicas: 1
  selector: { matchLabels: { app: myapp } }
  template:
    metadata: { labels: { app: myapp } }
    spec:
      containers:
      - name: myapp
        image: <acr>.azurecr.io/myapp:day53
        ports: [{ containerPort: 3000 }]
        resources:
          requests: { cpu: 50m, memory: 64Mi }
---
apiVersion: v1
kind: Service
metadata: { name: myapp, namespace: workloads }
spec:
  selector: { app: myapp }
  ports: [{ port: 80, targetPort: 3000 }]
  type: ClusterIP
```

## Done checklist

- Can map Service → endpoints → Pod without calling the Pod 'the Deployment'
- Manifests target namespace workloads, not kube-system
- Labels agree; port agrees; image tag is pinned
- Applied on kind/minikube OR traced on paper — AKS not created today

## Tomorrow — Day 56

AKS setup — optional gym membership with a cancel date